

RESEARCH SKILLS

WEEK 1: INTRODUCTION TO RESEARCH SKILLS

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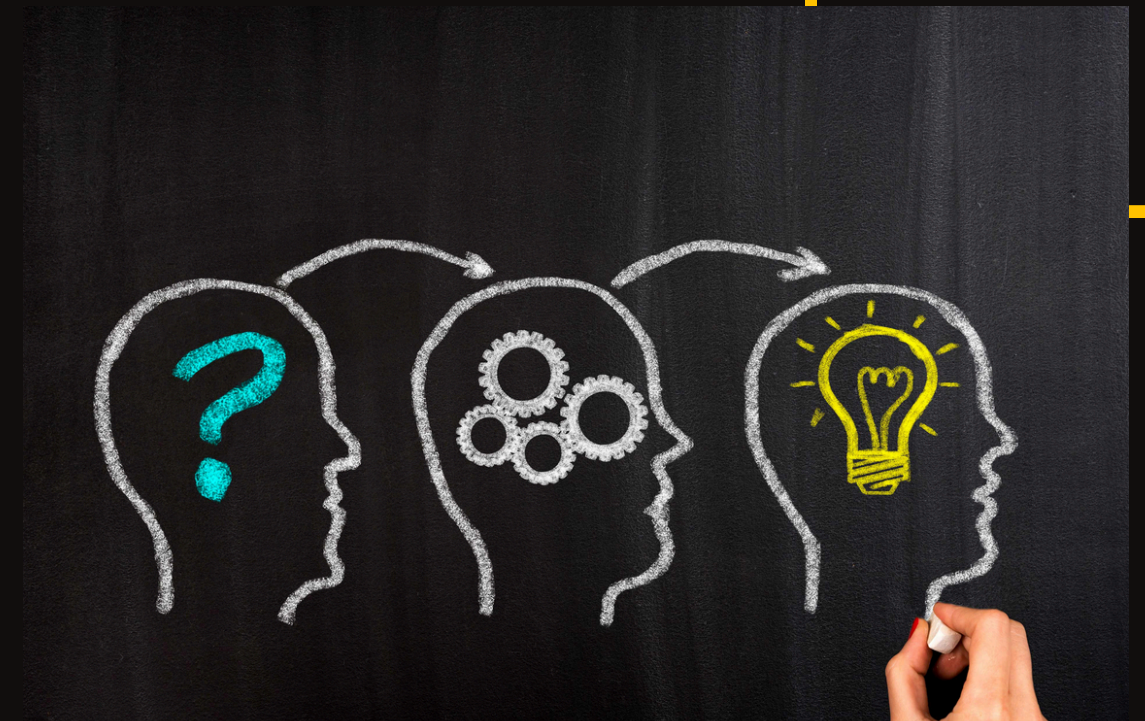
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The Research Cycle/Process



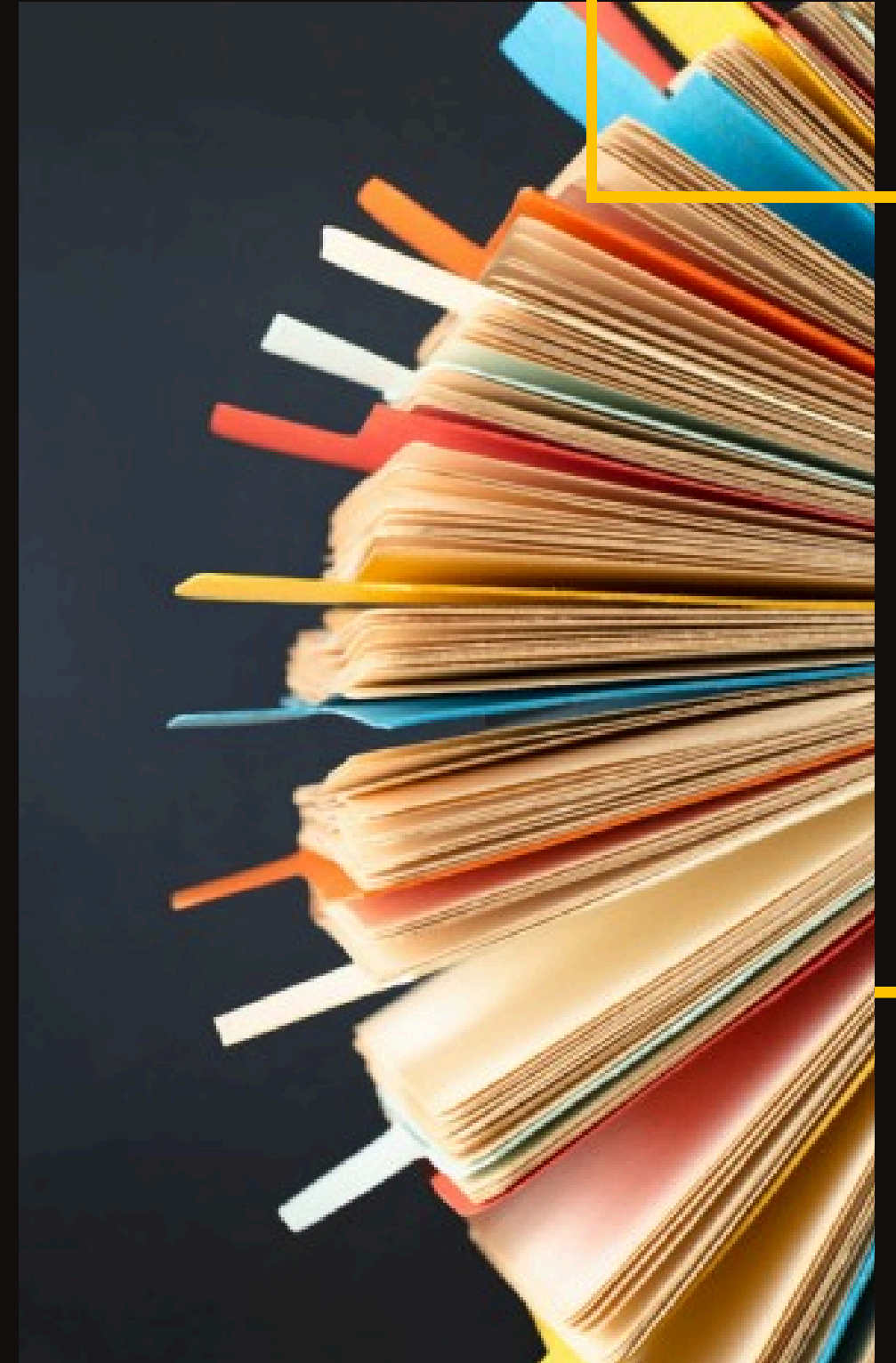
By the end of this session, students should be able to:

- Understand the concept and purpose of research.
- Recognize the key characteristics of scientific research.
- Explain the basic elements and steps involved in the research process.
- Appreciate the role of research in academic and real-world contexts.

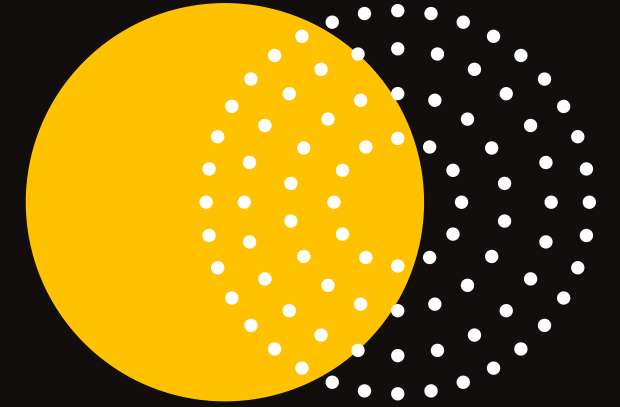


MAIN REFERENCES OF THIS COURSE

- Bell, J., & Waters, S. (2024). Doing Your Research Project: A Guide for First-Time Researchers (8th ed.). Open University Press.
- Creswell, J.W., & Creswell, J.D. (2023). Research Design: Qualitative, Quantitative, and Mixed Methods Approaches (6th ed.). SAGE Publications.
- Kumar, R. (2023). Research Methodology: A Step-by-Step Guide for Beginners (6th ed.). SAGE Publications.
- Bordens, K.S., & Abbott, B.B. (2022). Research Design and Methods: A Process Approach (11th ed.). McGraw-Hill Education.
- Leedy, P.D., & Ormrod, J.E. (2021). Practical Research: Planning and Design (12th ed.). Pearson Education.



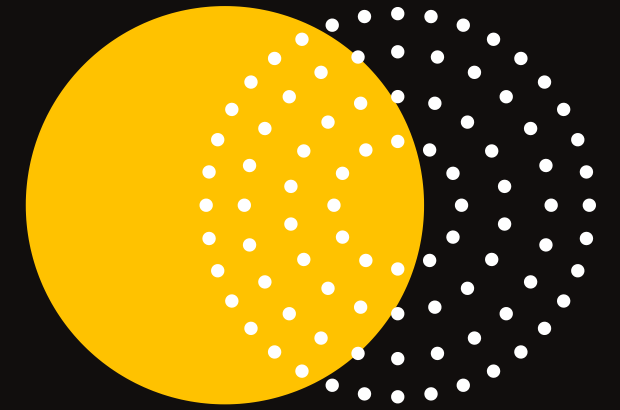
WHAT IS RESEARCH?



"Research is a systematic process of collecting, analyzing, and interpreting information to answer questions, solve problems, and generate new knowledge." — **Kumar, 2023**

"Research is a process of steps used to collect and analyze information to increase our understanding of a topic or issue." — **Creswell & Creswell, 2023**

WHAT IS RESEARCH?



Definition

Research is a systematic investigation into a study of materials, sources, or facts to establish new conclusions.

- ✓ Involves inquiry, investigation, and discovery
- ✓ Driven by questions or problems needing solutions

Core Purpose:

To create, validate, or refine knowledge using structured methods.

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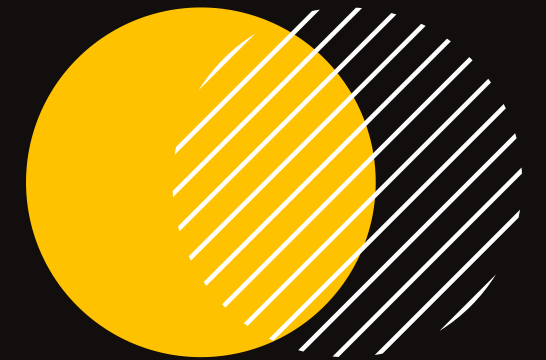
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-
- A magnifying glass with a black handle and frame is positioned over a newspaper page. The word "Research" is prominently displayed in a large, bold, black serif font within the lens of the magnifying glass. The background shows various elements of a newspaper: at the top, the word "DATE" is partially visible in large letters; below it, a dark horizontal bar contains the text "ics • Science • Health • Wellness • S"; further down, a light blue horizontal bar contains the text "The U"; and at the bottom, the word "Research" is repeated in a smaller font. To the right of the magnifying glass, there is a column of text that is partially obscured but includes words like "World's", "day to discuss", "warning and terrorism", "sues. Among other things", "will also discuss new", "sures on global security", "Last time this meeting", "very productive and", "brought major changes", "Earth. We will visit", "places of strategic interest", "and will discuss", "collaborations". The overall image conveys a sense of focused investigation or research.

KEY CHARACTERISTICS OF GOOD RESEARCH

(Adapted from Kumar, 2023; Bordens & Abbott, 2022)

- ✓ **Systematic** – follows an organized research process.
- ✓ **Logical** – based on clear reasoning and evidence.
- ✓ **Empirical** – relies on observable and measurable data.
- ✓ **Objective** – minimizes personal bias.
- ✓ **Reliable and Valid** – produces trustworthy and accurate results.
- ✓ **Replicable** – can be repeated by other researchers.
- ✓ **Transparent** – clearly reports methods and procedures.
- ✓ **Ethical** – respects ethical principles throughout the study.



BASIC ELEMENTS OF RESEARCH

Element	Description
Problem Statement	What you want to investigate.
Literature Review	What others have said about it.
Research Questions	Specific questions to answer.
Hypothesis (if applicable)	Prediction about the result.
Methodology	How you will conduct the research.
Data Collection	Tools/instruments for gathering info.
Data Analysis	How the data will be interpreted.
Conclusion	What you found out and its meaning.

THE RESEARCH CYCLE/PROCESS



EXAMPLE

Problem :Many university students experience stress during examinations.



Research Question :Does sleep duration affect students' exam performance?



Method :Survey 100 students.



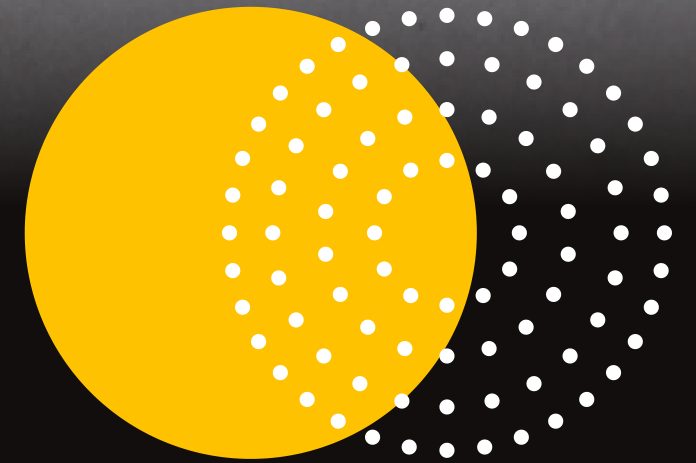
Results : Students sleeping more than 7 hours scored higher.



Conclusion : Adequate sleep improves academic performance.



DANKU!
THANK YOU!
MERCI!
GRAZIE!
GRACIAS!
DANK JE WEL!



THANK YOU